Rajarshi Chattopadhyay

Software Architect | AI Systems & Business Applications
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SUMMARY

Software architect and Staff Software Engineer with about 10 years of experience (IBM 2014–2018; Tesla 2020–present) designing and building production-grade intelligent systems for analytics platforms and business-critical applications. Focus on trustworthy AI agents that turn natural-language questions into verified SQL/DAX queries, secure data access (row-level security, injection-safe query generation), self-service data-engineering platforms, and full-stack finance and analytics applications. M.S. in Computer Science (Intelligent Systems), UT Dallas.

SKILLS

Architecture & Design: System design, distributed systems, data modeling, semantic layers, API design, OOP, security architecture
AI & Data: LLM agents, natural-language-to-SQL/DAX, prompt engineering, semantic search (vector + BM25), NLP, machine learning, deep learning (Keras, TensorFlow, PyTorch, scikit-learn)
Data Engineering: Apache Airflow, ETL/data pipelines, data warehouse/lakehouse, Spark, Hadoop, Kafka, Elastic stack
Languages & Web: Python, SQL, DAX, JavaScript/TypeScript, React, Java, Scala, Spring Boot, RESTful microservices
Data Stores: SQL, NoSQL, Elasticsearch, graph databases
DevOps & Cloud: Git, CI/CD (Jenkins), Docker, Kubernetes, UNIX/Linux, AWS (S3, EC2, EMR)

PROFESSIONAL EXPERIENCE

Tesla Dallas, TX

Staff Software Engineer *Feb 2026 – Present*

- Architect AI agents that generate, run, and explain analytical SQL/DAX queries from natural-language questions over enterprise financial data, replacing manual chart and report building.
- Design secure data access for external applications and agentic workflows: SQL-injection prevention, row-level security, and token-based access.
- Built hybrid semantic search (vector search plus BM25) to improve search relevance across an internal analytics platform.

Senior Software Development Engineer *Mar 2023 – Feb 2026*

- Sole owner of a visual, self-service data-engineering platform built on Apache Airflow; led it from requirements and architecture through development to MVP (Sep 2024) and launch (Jan 2025). Features include a visual DAG and task builder, test runs, Git integration with conflict resolution, team workspaces, and staging-to-production promotion. Became the standard pipeline-authoring front end for the data-engineering backend.
- Sole owner of a financial planning and analytics application giving cost-center leaders operating-expense (OPEX) visibility, with loan, lease, and delivery simulators; optimized query performance and resource usage.
- Built version 1.0 of an enterprise semantic layer and data modeler (SQL query-builder parser), the foundation for a single enterprise data model and business KPI layer.
- Developed the AI agent for charts and dashboards, workbook life cycle, and datasource editor of an internal analytics platform.
- Co-authored and presented a data lakehouse architecture proposal (2024).
- Designed, developed, and delivered a regulatory credit trading application that moved a revenue-critical department from spreadsheets to software, with contract and invoice generation, contract-lifecycle integration, and API integrations with other internal applications.
- Designed, developed, and delivered an application to productionize machine-learning model discovery and documentation, part of the infrastructure that democratizes data-science work across organizations.

Software Engineer *Mar 2021 – Feb 2023*

- Designed, developed, and deployed a company-wide metadata explorer, improving data discovery and lineage tracking from business-process assets to physical database entities.
- Set up the security architecture and deployment pipelines for cross-domain applications.

Data Engineer Intern *Fremont, CA | Jun 2020 – Nov 2020*

- Developed and automated ETLs for Enterprise Data Warehouse solutions for Sales, Delivery, and Logistics teams in a time-constrained environment using open-source technologies.

Software Engineer Intern, IT — Tesla Fremont, CA | Sep 2019 – Dec 2019

- Automated the handling and monitoring of secured infrastructure-access requests, cutting turnaround time by 90%, in support of the company's digital transformation.

Software Development Engineer Intern — Copart Dallas, TX | May 2019 – Aug 2019

- Designed and built a scalable RESTful microservices system for cross-verification and reconciliation of vendor check payments with the bank, preventing fraud worth thousands of dollars each month.
- Developed and maintained web application modules and REST microservices for billing and payments operations; explored architecture feasibility and trade-offs and prepared design documents.

Software Systems Engineer — IBM Kolkata, India | Dec 2014 – Jul 2018

- Developed and supported enterprise systems (client application back end) for a global shipping operation (A.P. Moller-Maersk).
- Led a 15-member team for end-to-end Level 3 application support at 100% SLA, developing automations for database updates and application-queue monitoring.
- Built MQ reporting tools: a Java UI for one-click daily queue-status reports and Bash automation for daily queue monitoring, eliminating manual effort.
- Provided on-the-job training on application system architecture and databases; received the IBM Best Emerging Techie Award (A.P. Moller-Maersk Diamond Awards, 2016).

Earlier: Administrative Assistant (student worker), Office of Graduate Education, UT Dallas (2018–2019); Embedded Systems Engineer Apprentice, Hi-Q Solutions (2012–2013; FSMs, assembly, VHDL); engineering apprenticeships at Bharat Sanchar Nigam Ltd. and Siemens VAI Metal Technologies (2012–2013).

EDUCATION

M.S., Computer Science (Intelligent Systems) — The University of Texas at Dallas 2018 – 2020
CGPA 3.64. Coursework: Data Structures, Big Data, Machine Learning.

B.Tech., Electronics and Communication Engineering — West Bengal University of Technology, India 2014
Thesis: Design and Development of a RISC Processor based on Harvard Architecture.

SELECTED PROJECTS

AI agent for energy data (2026): Safe natural-language-to-SQL agent over public U.S. EIA electricity-generation data: questions become parameterized single SELECT statements (user text never copied into SQL), a checker blocks writes, comments, extra statements, and out-of-schema columns, and unanswerable questions are refused. Python, SQLite. likarajo.github.io/work/ai-agent-chatbot

Neural Machine Translation: LSTM seq2seq encoder-decoder translation model in Keras. github.com/likarajo/language_translation

COVID-19 Sentiment Analysis (2020): Sentiment analysis of COVID-19 tweets in R. github.com/likarajo/covid19Sentiments

Customer Targeting: Binary classification model trained with multiple classifiers to predict a customer's product interest from ad-click and search data. github.com/likarajo/customer_targeting

Sentiment Analysis and Visualization: Spark Streaming application for live tweets via the Twitter API, sentiment analysis with Stanford CoreNLP, and real-time visualization with ZooKeeper, Kafka, Elasticsearch, Logstash, Kibana. github.com/likarajo/twitter_sentiments

More: house price prediction (Spark on AWS EMR), movie-review sentiment (CNN/RNN with GloVe), temperature-anomaly mapping (NASA data), Elasticsearch search app, chatbot. Full list: likarajo.github.io/Projects

AWARDS

- IBM Best Emerging Techie Award, A.P. Moller-Maersk Diamond Awards, 2016.
- First Runner-Up, IETE Technical Paper Contest, 2014: 16-bit RISC processor on an FPGA; also presented at a national seminar on Advanced Microprocessor and VLSI Devices.

CERTIFICATIONS & BADGES

Digital badges: credly.com/users/likarajo

- Oracle Certified Associate, Java SE 8 Programmer (2017)
- Chegg Skills AI Prompt Engineering Certificate (2024)
- IBM: Big Data Foundations L1 (2017), L2 (2019); Hadoop Foundations L1 (2017), L2 (2019); Hadoop Programming L1 (2017); Hadoop Data Access L1 (2019); Hadoop Administration L1 (2019); Spark L1 (2019); Docker Essentials (2019); IBM Cloud Kubernetes Service (2019); Blockchain Essentials (2017); Enterprise Design Thinking Practitioner (2016)

ACTIVITIES & WRITING

- Mentor to interns and new engineers; 2013 talks on computer architecture (IETE seminar) and wireless telecom.
- HackForHumanity 2018 hackathon; member and volunteer, Graduate Student Assembly and ACM, UT Dallas.
- Technical articles on LinkedIn and likarajoblogs.wordpress.com (Docker, CI/CD, RISC design, COVID-19 sentiment).